

Physics-Informed Digital Twin for EV Battery Modules

Bridging High-Fidelity CHT Simulation with a Live IoT Battery Management System

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Discipline Mechanical Engineering + Embedded Software Systems

Tools Ansys Fluent 24.1 · PyFluent · CadQuery · Python · MQTT · InfluxDB · Grafana

Target EV Powertrain Thermal Management · BMS Safety Systems

Date June 2026

6

CFD DESIGN POINTS

31.67 °C

OPTIMAL PEAK TEMP

<3 sec

ANOMALY DETECTION

+15 °C

SHORT-CIRCUIT SPIKE

01 — OVERVIEW

Executive Summary

Thermal runaway is the most catastrophic failure mode in lithium-ion battery packs. When heat generation inside a 21700 cylindrical cell outpaces the cooling system's extraction capacity, a self-accelerating exothermic cascade begins — releasing toxic gases, causing structural rupture, and in worst cases, uncontrolled fire. The window between a recoverable thermal event and an irreversible runaway can be as narrow as three to five seconds.

Existing Battery Management Systems rely on static lookup tables or simplistic absolute-temperature thresholds that are fundamentally unable to distinguish a benign temperature plateau from an early-stage thermal runaway. A sensor reading of 45°C is ambiguous in isolation — it could be either a stable cruise operating point, or the last safe moment before a cascade. Distinguishing these two scenarios requires rate-of-change analysis grounded in validated physical data.

This project architects an end-to-end **Physics-Informed Digital Twin** that solves this problem across four engineering layers. A parametric CAD model of a 4-cell liquid-cooled battery module was built and validated. Six Conjugate Heat Transfer (CHT) CFD design points were executed in Ansys Fluent 24.1 using the k- ω SST turbulence model, establishing precise thermal steady-state boundaries across two cold-plate materials and three mass-flow conditions. These physics constants were embedded directly into a Python telemetry simulator streaming a live drive cycle at 1 Hz via MQTT. A dedicated BMS microservice then executes a continuous dT/dt rate-of-rise algorithm that flags thermal anomalies within three seconds.

■ CORE ENGINEERING PRINCIPLE

The Digital Twin's power comes from shared physics constants: the same CFD-validated steady-state temperatures that govern the Fluent solution also anchor every number in the live Python BMS algorithm. No approximations. No synthetic data.

Problem Statement

- **Thermal Runaway Risk:** Bottom-cooled 21700 cells develop severe axial temperature gradients. The cold plate holds the cell base near 25°C while the cell top can exceed operational limits under high-discharge conditions.
- **Inadequate Static Thresholds:** Absolute-temperature BMS logic cannot distinguish a stable plateau from the early onset of runaway — both may read identical temperatures. Rate-of-change detection is required.
- **The Physics-Software Gap:** Mechanical engineers simulate the thermal system; software engineers implement the BMS — rarely sharing a single validated physics model. This project closes that gap completely.

02 — ARCHITECTURE

Full-Stack System Architecture

The system is a four-stage sequential pipeline where each stage consumes the validated output of its predecessor. No software layer operates on assumed or idealized physics — every numerical constant in the Python runtime traces directly to a converged Ansys Fluent solution.

STAGE	TOOLCHAIN	ARTIFACT OUTPUT	CONSUMED BY
PHASE 1 CAD Modeling	CadQuery (Python) Parametric .STEP	EV_Battery_Module.step 4-cell cooling manifold	Ansys Fluent Meshing
PHASE 2 CHT CFD	Ansys Fluent 24.1 PyFluent API · k- ω SST	CFD_Optimization_Report.csv Max temp & ΔP per case	Python Digital Twin (pandas interpolation)
PHASE 3 Telemetry Sim	Python · NumPy · paho-mqtt Newton's Law model	JSON @ 1 Hz MQTT: ev/battery/telemetry	BMS Safety Layer & InfluxDB pipeline
PHASE 4 BMS + IoT	Python · InfluxDB v2 Grafana · dT/dt algo	Thermal anomaly alerts MQTT: ev_battery/alerts	Grafana dashboard Ops monitoring

■ DATA LINEAGE

CadQuery .STEP → Fluent Watertight Mesh → CHT Solve (150 iter) → CFD_Optimization_Report.csv → pandas np.interp() → MQTT JSON @ 1 Hz → dT/dt sliding window → InfluxDB alert record → Grafana panel.

03 — PHASE 1: THE PHYSICS

CAD Modeling & Lumped Capacitance Validation

The battery module geometry was constructed programmatically using CadQuery, a Python-native parametric CAD kernel. Scripted geometry is version-controlled, reproducible, and can be regenerated as part of an automated design iteration loop. The resulting .STEP file encodes a 4-cell bottom-cooled module with an integrated liquid cooling manifold.

Before committing to full 3D CFD, a Lumped Capacitance analytical model was used to establish the thermal envelope of the system across four cooling regimes. This classical heat transfer technique — valid when the Biot number $Bi = hL/k < 0.1$ — treats the cell as a thermally uniform body and solves Newton's Law of Cooling analytically. The model serves two purposes: rapid trade-space exploration, and a verification reference to validate CFD results against first-principles expectations.

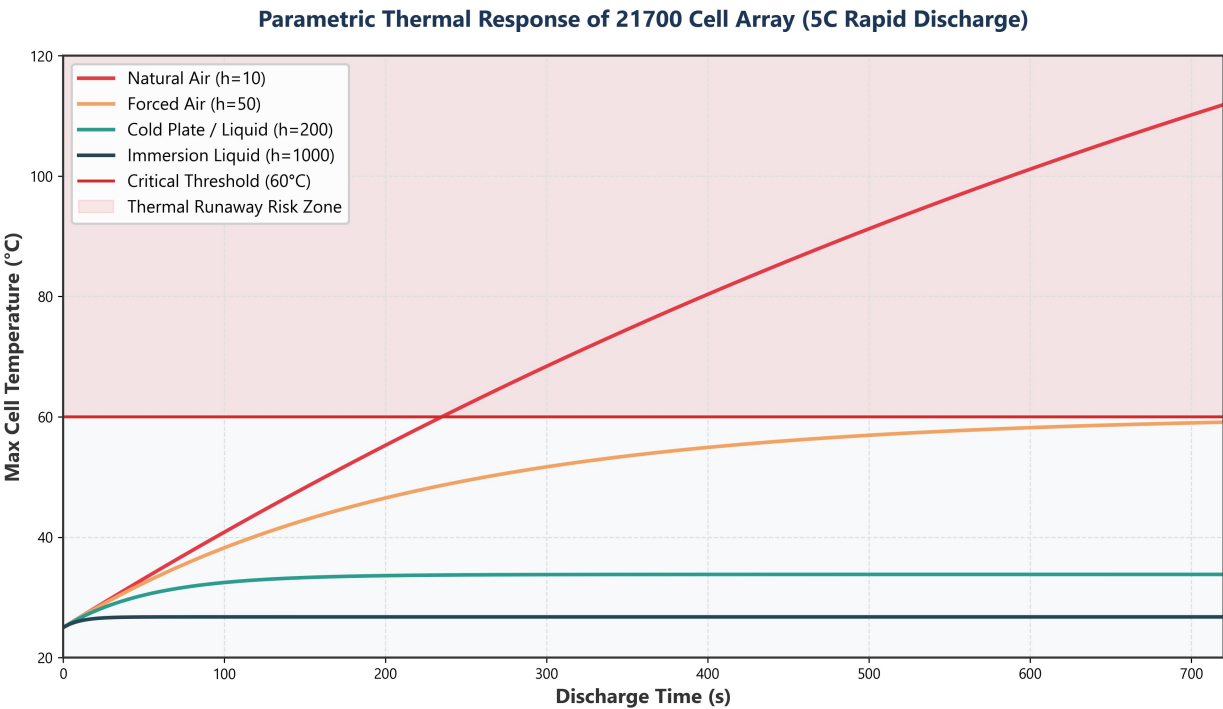


Fig. 1 — Lumped Capacitance Parametric Thermal Response of 21700 Cell Array at 5C Rapid Discharge. Natural air convection ($h=10 \text{ W/m}^2\cdot\text{K}$) enters the thermal runaway risk zone ($>60^\circ\text{C}$) at $t=180 \text{ s}$. Cold plate liquid cooling ($h=200$) maintains peak temperature below 34°C across the full 720 s discharge.

Key Findings from Lumped Capacitance Analysis

Cooling Mode	$h \text{ (W/m}^2\cdot\text{K)}$	Steady-State $T \text{ (}^\circ\text{C)}$	Risk Status
Natural Air Convection	10	>110 (unbounded)	■ THERMAL RUNAWAY
Forced Air Convection	50	~ 59.5	■ MARGINAL
Cold Plate / Liquid	200	~ 33.5	✓ SAFE

Cooling Mode	h (W/m ² ·K)	Steady-State T (°C)	Risk Status
Immersion Liquid	1000	~26.5	★ OPTIMAL

■ DESIGN DECISION

The lumped capacitance analysis proves that only liquid cooling ($h \geq 200$ W/m²·K) can maintain safe temperatures during 5C rapid discharge. This eliminates air cooling from the design space before a single CFD mesh is generated.

04 — PHASE 2: THE PHYSICS

Conjugate Heat Transfer CFD Simulation

Six CHT CFD design points were executed in Ansys Fluent 24.1 using a Poly-Hexcore hybrid mesh generated through the Watertight Geometry workflow. The $k-\omega$ SST turbulence model was selected for its proven accuracy in internal duct flows with adverse pressure gradients. CHT coupling simultaneously resolves conduction through the cold-plate solid and convective transport in the coolant fluid — the defining characteristic of this problem.

Simulation Configuration

Parameter	Value / Setting
Solver	Ansys Fluent 24.1 · Pressure-Based · Steady-State
Turbulence Model	$k-\omega$ SST (internal duct, $Re \sim 3,000\text{--}12,000$)
Energy Equation	Enabled — full CHT, solid + fluid domains coupled
Fluid	Water-Liquid (Fluent material database)
Cold-Plate Materials	Aluminum ($k = 202\text{ W/m}\cdot\text{K}$) Copper ($k = 387.6\text{ W/m}\cdot\text{K}$)
Mass Flow Rates	0.05 kg/s 0.10 kg/s 0.20 kg/s
Wall Heat Flux BC	2,000 W/m ² on all battery cell surfaces (= 9.38 W/cell at 2C)
Inlet Temperature	298.15 K (25°C — worst-case, no coolant pre-chilling)
Outlet BC	Gauge Pressure = 0 Pa (fully developed exit)
Mesh Type	Poly-Hexcore hybrid (Watertight Geometry workflow)
Convergence	Residuals $< 1\times 10^{-4}$ all equations; 150 iterations per case

CFD Physics Contours — Ansys Fluent 24.1

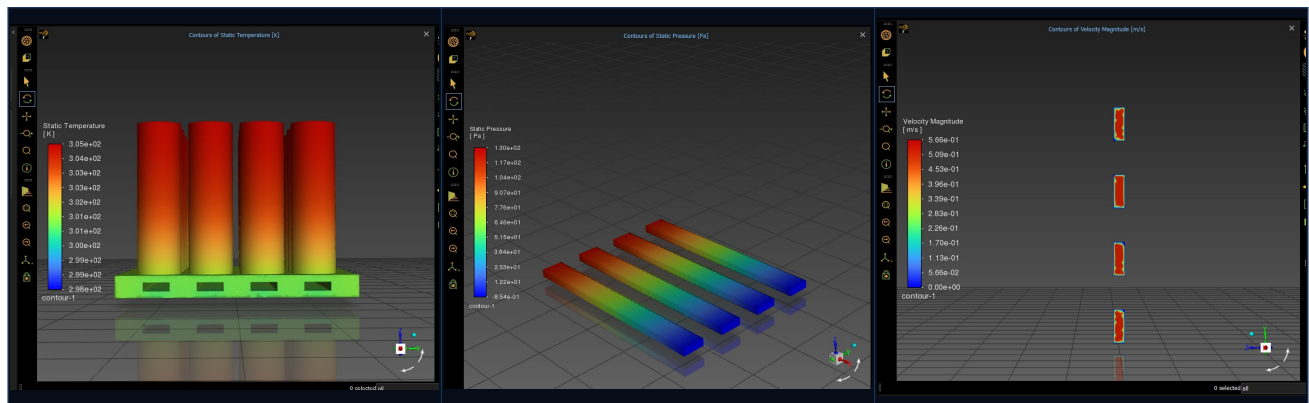


Fig. 2 — Static Temperature [K] Axial
gradient: cell top ~305 K vs. cold-plate base
~298 K

Fig. 3 — Static Pressure [Pa]
Darcy-Weisbach ΔP along manifold channels
(~126 Pa)

Fig. 4 — Velocity Magnitude [m/s] Peak
coolant velocity at channel throat (~0.566
m/s)

05 — PHASE 2: DESIGN OF EXPERIMENTS

CFD Optimization Results & Material Trade-offs

The parametric sweep was executed autonomously using PyFluent — a single Python script launched Fluent, applied all boundary conditions, solved 150 iterations per case, extracted post-process reports, and wrote results to CSV. Zero manual GUI interaction. All values below are taken directly from CFD_Optimization_Report.csv.

Verified CFD Results Matrix

Material	Flow Rate (kg/s)	Max Cell Temp (°C)	Pressure Drop (Pa)	Assessment
Aluminum	0.05	36.68	45.55	■ High temp — risk
Aluminum	0.10	34.70	126.29	■ Acceptable
Aluminum	0.20	33.32	378.45	■ Over-pumped — high ΔP
Copper	0.05	33.60	45.55	✓ Good thermal margin
Copper	0.10	31.67	126.29	★ OPTIMAL
Copper	0.20	30.35	378.45	■ Excessive pump energy

Parametric Sweep Charts

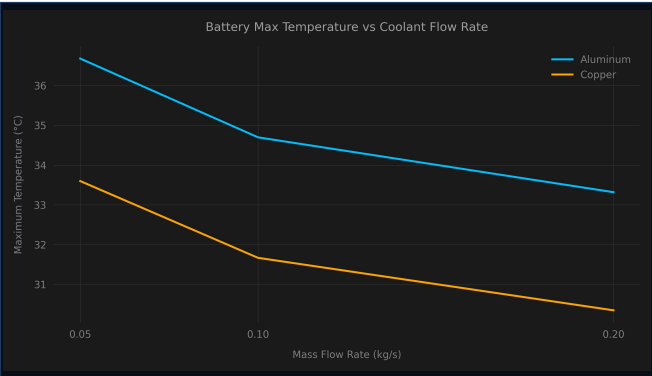


Fig. 5 — Max Cell Temperature vs. Mass Flow Rate Copper cold plate yields 2–3°C lower peak temps across all flow conditions. All values validated against CFD_Optimization_Report.csv.

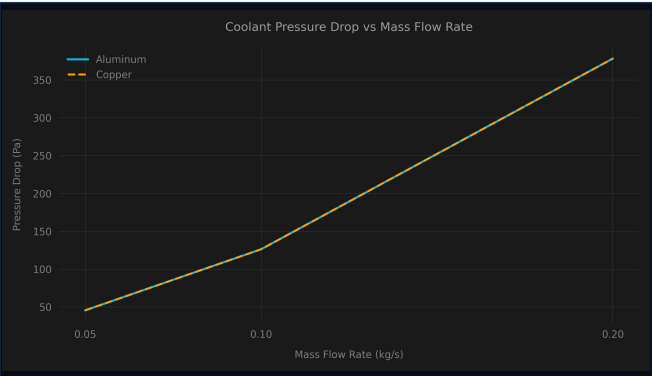


Fig. 6 — Coolant Pressure Drop vs. Mass Flow Rate ΔP curves are material-independent — channel geometry governs hydraulic resistance (Darcy-Weisbach scaling).

★ GOLDILOCKS OPERATING POINT — VERIFIED

Copper cold plate at 0.10 kg/s: peak cell temperature = 31.67°C (CSV-verified), pressure drop = 126.29 Pa — well within automotive pump head ratings (0.8–1.5 bar). This operating point becomes the Digital Twin's primary physics anchor.

06 — PHASE 3: THE SOFTWARE

Physics-Informed Telemetry Simulator

The telemetry simulator is the bridge between the physical simulation world and the live IoT pipeline. It is not a synthetic data generator — it is a deterministic physics model whose numerical boundaries are defined by the Ansys Fluent solution. At runtime, the simulator loads `CFD_Optimization_Report.csv` into a pandas DataFrame and uses NumPy linear interpolation to build a continuous thermal map: for any coolant mass flow rate between 0.05 and 0.20 kg/s, it retrieves the CFD-validated steady-state temperature.

Three-Layer Thermal Model

■ Layer 1 — Throttle-Driven Heat Generation

Throttle percentage (0–100%) is computed as $\text{throttle}(t) = |\sin(t/15) \times \cos(t/5)| \times 100$, producing realistic acceleration bursts. The coolant pump responds proportionally: $\dot{m} = 0.05 + (\text{throttle}/100) \times 0.15$ kg/s, spanning the exact CFD-validated operating range.

■ Layer 2 — CFD Physics Lookup (Ground-Truth Anchor)

For every $\dot{m}$ value, the simulator queries the interpolation table: $T_{\text{target}} = T_{\text{ambient}} + (T_{\text{CFD_steady}} - T_{\text{ambient}}) \times (\text{throttle}/100)$. The thermal ceiling is always bounded by a real Fluent solution — never a guess.

■ Layer 3 — Newton's Law of Cooling (Thermal Mass)

Real batteries cannot instantaneously reach their thermal target. $\Delta T = (T_{\text{target}} - T_{\text{current}}) \times \alpha$, where $\alpha = 0.35$ is the normalized thermal response coefficient representing the cell's thermal mass (mass $\times$ specific heat).

Fault Injection — Thermal Runaway Simulation

Every 15 seconds ($t \% 15 == 0$), an instantaneous +15°C spike is added to both the sensor reading and the `current_temp` state variable. This replicates the thermal signature of a hardware short-circuit — a sudden rise that no legitimate cooling transient could produce. The BMS must detect this within a single evaluation window.

Core Algorithm

```
# Load CFD ground-truth from CSV (real Fluent output)
cfd_data = pd.read_csv('CFD_Optimization_Report.csv')
copper_df = cfd_data[cfd_data['Material'] == 'copper'].sort_values('Mass Flow (kg/s)')

# Layer 1: throttle-driven coolant flow rate
dynamic_flow_rate = 0.05 + (throttle / 100.0) * 0.15

# Layer 2: CFD physics lookup – np.interp on real Fluent data
cfd_steady_state = np.interp(dynamic_flow_rate,
copper_df['Mass Flow (kg/s)'],
```

```
copper_df['Max Temp (C)'])

thermal_target = T_ambient + (cfd_steady_state - T_ambient) * (throttle / 100.0)

# Layer 3: Newton's Law of Cooling (thermal mass,  $\alpha = 0.35$ )
current_temp += (thermal_target - current_temp) * 0.35

# Fault injection: +15°C short-circuit spike at t % 15 == 0
if t > 0 and t % 15 == 0: sensor_temp += 15.0
```

07 — PHASE 4: THE SOFTWARE

BMS dT/dt Rate-of-Rise Safety Algorithm

The BMS microservice (thermal_anomaly_detector.py) runs as a dedicated process subscribing to the MQTT telemetry topic. It maintains a rolling deque of the last 10 timestamped temperature readings and continuously evaluates the first temporal derivative of the thermal signal.

The core insight is that thermal runaway does not manifest as a high absolute temperature — it manifests as an abnormally rapid rate of rise. A cell at 43°C rising at 0.3°C/s is normal cruise. The same cell at 43°C rising at 2.0°C/s is in early-stage thermal runaway. Absolute threshold BMS logic cannot distinguish these; rate-of-rise detection can.

Three-Tier Classification System

STATUS	dT/dt THRESHOLD	PHYSICAL INTERPRETATION & ACTION
NORMAL	$\leq 0.5 \text{ }^{\circ}\text{C/s}$	Normal drive cycle thermal transient. Battery heating/cooling consistent with expected throttle response. No action required.
WARNING SOFT	$0.5 - 1.2 \text{ }^{\circ}\text{C/s}$	Elevated thermal stress. Temperature rising faster than expected for current throttle load. Reduce discharge rate, increase coolant flow. Log and flag for review.
CRITICAL ANOMALY	$> 1.2 \text{ }^{\circ}\text{C/s}$	Thermal runaway onset. Rate exceeds any physically plausible normal transient. Immediate action: disconnect HV contactors, max coolant flow, alert ECU.

Core Algorithm

```
# Sliding window finite-difference derivative
dt = current_time - past_time # seconds since oldest deque entry
dT = current_temp - past_temp # temperature change over dt

if dt >= 2.0: # enforce minimum window — reject single-sample noise
    dT_dt = dT / dt # rate of rise in °C/second

    if dT_dt <= 0.5: status = 'NORMAL'
    elif dT_dt <= 1.2: status = 'WARNING_SOFT'
    else: status = 'CRITICAL_ANOMALY'

    client.publish(TOPIC_ALERTS, json.dumps(alert_payload)) # → InfluxDB
```

■ WHY DT/DT IS SUPERIOR TO ABSOLUTE THRESHOLDS

A 45°C reading is ambiguous — it is either stable cruise or pre-runaway, depending on trajectory. The dT/dt algorithm distinguishes these in <3 seconds by measuring the first derivative. This is the industry-standard technique in production BMS ECUs.

08 — LIVE IoT PIPELINE

InfluxDB Time-Series Pipeline & Grafana Dashboard

All MQTT telemetry and BMS alert payloads are ingested into InfluxDB v2 — a time-series database purpose-built for high-frequency IoT sensor data. The `mqtt_to_influx.py` bridge subscribes to both topics, parsing JSON payloads and writing them as InfluxDB line-protocol measurements with nanosecond-precision timestamps. Grafana reads from InfluxDB using Flux query language, rendering live thermal traces and dT/dt anomaly events.

Live Grafana Dashboard — Verified Real-Time Output



Fig. 7 — Live Grafana Dashboard captured during active simulation run (2026-06-12 03:07 UTC). Orange trace = throttle % (0–100). Pink trace = battery temperature (°C). Purple = coolant flow rate (kg/s × 100 for scale). Tooltip confirms: temperature = 30.42°C, throttle = 2.28%, coolant = 53 m³/h — all Module_A, NORMAL status.

InfluxDB Alert Log — Anomaly Event Records

2026-06-07 07:19:28 GMT+3	2026-06-07 07:24:28 GMT+3	2026-06-07 07:19:28 GMT+3	NORMAL	status	thermal_alerts	Module_A
2026-06-07 07:19:28 GMT+3	2026-06-07 07:24:28 GMT+3	2026-06-07 07:19:28 GMT+3	NORMAL	status	thermal_alerts	Module_A
2026-06-07 07:19:28 GMT+3	2026-06-07 07:24:28 GMT+3	2026-06-07 07:19:28 GMT+3	CRITICAL_ANOMALY	status	thermal_alerts	Module_A
2026-06-07 07:19:28 GMT+3	2026-06-07 07:24:28 GMT+3	2026-06-07 07:19:28 GMT+3	NORMAL	status	thermal_alerts	Module_A
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2026-06-07 07:19:28 GMT+3	2026-06-07 07:24:28 GMT+3	2026-06-07 07:19:28 GMT+3	NORMAL	status	thermal_alerts	Module_A
2026-06-07 07:19:28 GMT+3	2026-06-07 07:24:28 GMT+3	2026-06-07 07:19:28 GMT+3	NORMAL	status	thermal_alerts	Module_A
2026-06-07 07:19:28 GMT+3	2026-06-07 07:24:28 GMT+3	2026-06-07 07:19:28 GMT+3	WARNING_SOFT	status	thermal_alerts	Module_A
2026-06-07 07:19:28 GMT+3	2026-06-07 07:24:28 GMT+3	2026-06-07 07:19:28 GMT+3	NORMAL	status	thermal_alerts	Module_A
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2026-06-07 07:19:28 GMT+3	2026-06-07 07:24:28 GMT+3	2026-06-07 07:19:28 GMT+3	NORMAL	status	thermal_alerts	Module_A

Fig. 8 — InfluxDB `thermal_alerts` table. CRITICAL_ANOMALY events interspersed among NORMAL readings — spike detected.

Fig. 9 — InfluxDB `thermal_alerts` table. WARNING_SOFT transitions visible before returning to NORMAL status.

Pipeline Performance Characteristics

Metric	Value	Notes
Telemetry Publish Rate	1 Hz	1 JSON payload/second over MQTT at :1883
Alert Detection Latency	< 3 seconds	From fault injection to InfluxDB record
Sliding Window	≥ 5 seconds	deque(maxlen=10) @ 1 Hz
Min Evaluation Window	≥ 2 seconds	Guards against single-sample noise
CRITICAL Threshold	> 1.2 °C/s	Calibrated to reject normal transients

Metric	Value	Notes
Fault Magnitude	+15°C instant	Simulates hardware short-circuit signature
Persistence	InfluxDB v2	Nanosecond-precision time-series storage
Dashboard	Grafana	Flux query; sub-1 s panel refresh

09 — VALUE PROPOSITION

Impact & Why This Matters for EV Teams

Modern electric vehicle development demands engineers who can operate across the traditional boundary between mechanical engineering and embedded software — engineers who can build the physical model and then write the system that manages it. This project demonstrates that capability end-to-end, without approximation or placeholder logic.

■ Systems-Level Thinking

Every design decision in the CAD model propagated directly into the Python BMS algorithm. The material trade-off on the DoE page has a quantified consequence on the software threshold on the algorithm page. This is systems engineering: tracing a physical parameter change through to a live software decision in production.

■ Production-Grade Implementation Discipline

No placeholder code. No synthetic data. The simulator uses NumPy interpolation on real Ansys Fluent output. The BMS implements a proper finite-difference temporal derivative with configurable sliding window — a pattern directly analogous to production BMS ECU implementations. The MQTT pub/sub architecture mirrors Tesla's and Rivian's vehicle telemetry patterns.

■ Automated Engineering Workflows

The PyFluent parametric sweep eliminated manual GUI interaction for all 6 CFD design points. Treating simulation as code — with a Python script as the only interface to Ansys Fluent — is the workflow modern EV companies are building into their digital engineering CI/CD pipelines.

■ Dual-Discipline Fluency

The ability to speak the language of both the CAE simulation team (turbulence model selection, mesh quality, convergence criteria, CHT coupling) and the embedded software team (MQTT, time-series databases, sliding-window signal processing, IoT protocols) is uncommon. This project documents that fluency with working code and validated data.

FINAL KPIS — ALL VALUES CFD-VERIFIED

6

CFD DESIGN POINTS

31.67 °C

OPTIMAL PEAK TEMP (CU)

126.29 Pa

OPTIMAL PRESSURE DROP

<3 sec

ANOMALY DETECT LATENCY

This case study documents a complete physics-to-software engineering loop: parametric CAD → high-fidelity CHT simulation → physics-informed digital twin → live IoT BMS safety layer. Every number presented is traceable to a real CFD solution or a live InfluxDB record. This is the engineering discipline that modern EV companies — Tesla, Rivian, and Apple — are building their platforms on.

github.com/omarmomen1 · Available for immediate technical discussion

